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MILITARY SPECIFICATION

MOWERS, LAWN; FLAIL TYPE, TOWED AND TRACTOR MOUNTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers independently powered, and power takeoff (pto) powered, towed and tractor mounting flail type mowers.

1.2 Classification. Towed and tractor mounted flail type mowers shall be of the following types, styles and nominal sizes, as specified (see 6.2):

Type I - Towed, gasoline engine powered.

Style A - Single-unit 48-inch swath.
Style C - 3-unit-gang 11-foot 6-inch swath.
Style D - 5-unit-gang 19-foot swath.

Type III - Rear tractor mounting (three point hitch), pto powered.

Size 1 - 48-inch swath.
Size 2 - 60-inch swath.
Size 3 - 72-inch swath.

Type V - Sidearm, tractor mounting, pto powered.

Size 3 - 72-inch swath.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3750

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81 - Tags, Shipping and Stock.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproof, Flexible.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-T-704 - Treatment and Painting of Material.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-P-46093 - Primer Coating, Synthetic (For Brake Drums).
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E18 - Standard for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF AGRICULTURAL ENGINEERS (ASAE)

S211.3 - Standard for V-Belt Drives for Farm Machines.

(Application for copies should be addressed to the American Society of Agricultural Engineers, 420 Main Street, St. Joseph, MI 49085.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE HANDBOOK

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Assocition, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The mowers shall be flail type. Type I mowers shall be independently gasoline-engine-powered, and towed singly, or in groups of three or five in a towing frame with manually set cutter elevation for transport. Type III mowers shall be pto powered, and tractor rear-end mounted by a three point hitch, with elevation transport control by the tractor operator. Type V mowers shall be pto powered, and tractor side mounted, with hydraulic elevation transport control by the tractor operator. Means for height of cut adjustments shall be provided for all types.

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3.2 First article. When specified (see 6.2), the contractor shall furnish a mower of the type, style and size as required for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. Each mower shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in each mower being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All mowers of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories and spare parts. No deviation will be acceptable without prior written approval of the contracting officer.

* 3.6 Ease of maintenance. The mowers shall be so designed and constructed that normal adjustments, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or disturbance of other elements of the mower. Ease of maintenance provisions shall insure operating clearances for facilitating maintenance and servicing. Where possible, intricate locking devices, controls and threaded fastenings that can be easily over-torqued by operators lacking feel through thick gloves or numbness, shall be avoided. Covers or plates that must be removed for component adjustment or for parts removal shall be equipped with substantial, quick-disconnect fastenings.

* 3.7 Safety. All rotating and reciprocating parts, and all parts subject to high operational temperatures, that are of such a nature or so located as to be or become a hazard to the safety of the operating personnel shall be insulated, enclosed or guarded to the extent necessary to eliminate the hazard. The mower shall comply with applicable OSHA regulations in effect at time of manufacture. In addition, type III and type V mowers shall meet the safety criteria for industrial flail mowers of SAE J100.

3.8 Performance. When the engine is operating at its rated speed, the type I mower shall display no evidence of dynamic imbalance. Cutting shall be effective at towing travel speeds up to 4 miles per hour. Mowing at maximum

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towing speeds over rough terrain through tight turns shall not result in yawing and skipping. At rated travel speeds, type III and type V mowers shall track smoothly without bounce. The axle centerline of the cutting assembly shall travel parallel to the travel surface. Each swath shall be cut with substantially no grass, weed or brush remaining above the height of cut for which the mower is set. The width of cut shall be not less than that applicable to the style or size of mower furnished. Random growths of grass, briars, weeds and bushes not less than 10 inches deep shall be cut to even height and width. Collision of the cutting assembly with solid objects, such as bottles, bricks and small boards, shall not impede mowing and cause ejection of solid particles to either side; not to more than 4 feet forward or rearward from the cutting assembly.

* 3.9 Design and construction. All structural and wearing parts of the mower shall be designed for continuous rugged service. Drive line shaft and universal joints shall be shielded. All other moving parts, such as the cutting assembly, shafts and belt drives, shall be shielded as required for safety and protection from damage due to the entrance of dust, water and foreign objects. Power for type I mowers shall be distributed from a centrally located engine or gearbox mounted atop the cutting assembly housing, and transmitted through a ball or a roller bearing supported drive shaft and C-section V-belt drive. Belt drives shall meet or exceed the requirements of ASAE S211.3, as applicable to the pulley arrangement and the power transmitted. Type I power trains shall be designed for continuous transmission of not less than 20 horserpower (hp). Type III and type V power trains shall be designed to transmit the pto drive power output continuously from the tractors rated at not less than 60 hp. Type III and type V gearing ratios shall be designed to obtain effective cutting knife peripheral speed when powered from continuous running pto drives of either 540 revolutions per minute (rpm) in accordance wih SAE J718, or 1,000 rpm in accordance with SAE J719, as specified (see 6.2), with independent pto drive in accordance with SAE J722.

3.9.1 Cutting assembly. The cutting assembly shall consist of a belt driven steel shaft or drum, with replaceable free swinging self-cleaning flail knives. Unless otherwise specified (see 6.2), fine cutting knives shall be provided. When specified (see 6.2), the manufacturer's heavy duty coarse cutting knives shall be provided. Knife deflection at effective mowing speeds shall be adequate for safety, and the prevention of disabling damage. Knife hardness shall range between Rockwell hardness C43 and C47. The completely assembled cutting assembly shall be dynamically balanced when rotating at the speed required for effective cutting. A protective housing, designed to prevent the buildup of damp clippings on the underside of the housing surrounding the cutting assembly, shaped and placed to prevent dangerous ejection of solid objects, shall be provided.

3.9.2 Flotation. The type I, style A mower unit shall float on a full length roller, behind the cutting assembly, and ball or roller bearing mounted, pneumatic-tired wheels at each end of the cutting assembly. The wheels shall be mounted with sufficient flexibility to prevent damage when traveling over bumps and holes, and adequate rigidity to maintain the set height of cut. The type III mower shall be flexibly mounted on the tractor three point hitch lifting assembly. The type V mower shall be supported by the raising arm on the tractor. Type I shall have a height of cut adjustment range between 3/4 inch and not less than 6 inches. The height of cut adjustment range for types III and V shall be between 1 inch, and not less

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than 6 inches. Towing devices shall be ball and coupler type conforming to SAE J684, a commercial pin and clevis type with adapters for use with commercial tractors, or a lunette ring of 1-5/8 inch diameter steel stock with a 3-inch diameter towing eye, as specified (see 6.2).

3.9.3 Transportability. The type I, style A mower shall be provided with an adjustment position of the cutting assembly to prevent any part from striking the corner of the curb when towed over a 4-inch high curb with the roller parallel to the curb. Mowers without wheels of adequate size for transport towing shall have a centrally located lifting attachment suitable for transport in level position on the hook of a service truck crane. Style C and D mowers not designed for transport by towing shall be provided with frame designed to transport the gang in level position by service truck crane. The frame may be either an accessory, or an independently wheel-supported part of the gang mower assembly. The type III mower shall be transported without damage by elevating mechanism of the three point hitch, at maximum speeds applicable to the tractor. The type V mower shall be transported without damage and loss of lubricants when secured with the lifting arm in vertical position at maximum tractor speeds.

3.9.4 Tires. Pneumatic or semi-pneumatic type tires shall be the manufacturer's first line grade and shall have the individual rated load-carrying capacities equal to the maximum tire loading imposed by the operation. When tube type tires are furnished, tubes shall be of heavy-duty type and shall be of proper size. Tires, rims and loadings shall be in accordance with the TRA recommendations.

3.10 Leaf mulcher. When specified (see 6.2), a leaf mulching attachment, for thoroughly shredding or pulverizing leaves, shall be provided.

3.11 Type I, style A mower unit. The mower unit shall have a cutting assembly designed to cut a 48-inch swath at each pass, and shall be integrally assembled, with the gasoline engine mounted on the cutting assembly housing. Each mower unit shall have a towing arm, and housing-mounted ball or universal type arm attachment devices, placed to provide for alternative V-formation style C and D mowing, or for towing in tandem transport formation. Mower units in style C and D fleet formations shall be interchangeable.

* 3.11.1 Gasoline engine. The gasoline engine shall be a four cycle standard commercial design that has been proved to be satisfactory in extensive use and shall be one for which repair parts are readily obtainable. The power and speed required for the mower shall not exceed the continuous hp value at which the engine is rated. The engine shall operate satisfactorily with minimum oil level when positioned at any angle between horizontal and 15 degrees (°) from horizontal. The engine shall be equipped with manual starting means, and efficient dry type air cleaner, a spark-arresting muffler, an adjustable hand throttle, a choke, a governor and a resistor type spark plug. The cooling system shall be such that the engine will operate continuously in ambient temperatures between 32° and 100° Fahrenheit. The lubricating systems shall provide adequate lubrication under all operating conditions, and shall include a dipstick oil level indicator or overflow oil level filler spout, and a conveniently located oil reservoir drain plug. A strainer shall be provided in the fuel supply line and an efficient carburetor shall be furnished. A fuel tank with sufficient capacity for not less than

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4 hours normal operation shall be furnished. The gasoline engine shall be centrally mounted on the cutting assembly housing and connected through a flexible coupling and belt drive train to the cutting assembly. Vibration isolation protection for the engine shall be provided. The engine shall meet the requirements specified herein using gasoline that is unleaded, or having a low-lead content. Low-lead gasoline shall not contain more than 0.5 gram of tetraethyl lead per gallon of gasoline.

3.12 Type III, rear tractor mounting type. Rear tractor mounting mower shall be attached to the tractor with a three-point hitch linkage assembly in accordance with SAE J715, and a shielded telescoping pto drive shaft in accordance with SAE J721. The drive train shall be shock protected, and geared for the appropriate pto drive speed required to operate the cutting assembly at the effective cutting speed. The implement attachment assembly of the three-point hitch shall not interfere with mower flotation during operation on ground contours.

3.13 Type V, sidearm, tractor mounting type. The type V mower shall be fitted for side mounting on a high-speed, low center of gravity type tractor of specific make and model, as specified (see 6.2). The mower shall be equipped with a free end protective guide skid and a full length roller. The hinge pto drive mounting end shall have safety breakaway protection, and be placed to provide for operation with the mower resting by gravity on slopes between 45° below, and 60° above the horizontal level of tractor operation. The drive train and gear reduction shall be compatible with the tractor furnished, for effecive cutting assembly speed.

3.14 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

* 3.15 Cleaning, treatment and painting. Unless otherwise specified (see 6.2), surfaces normally painted in good commercial practice shall be cleaned, treated and painted as specified herein. When specified (see 6.2), the unit shall be cleaned, treated and painted in accordance with MIL-T-704, type A or C. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not

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less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel or other defects.

3.16 Instruction plates. Each mower shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment.

* 3.17 Workmanship.

3.17.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.17.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts and screws shall be tight.

3.17.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the member.

3.17.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.17.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on each mower when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction and manufacturing technique applicable to the remaining mowers to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.2, and the packaging inspection of 4.5.

4.3 Examination. Each mower shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article shall be tested in accordance with 4.4.1. Production mowers shall be tested in accordance with 4.4.2.

4.4.1 First article tests. The first article shall be towed or tractor mounted as applicable to the type, and shall be tested by operation over uncleared land covered with an average stand not less than 10 inches deep of grass, weeds, briars and light brush. The test site for type V shall include a location with flat sloped surfaces at angles up to not less than 60° above level surface, and down to not less than 45° below level surface. The first article shall be operated for not less than 1 hour in the following order: 20 minutes at 6-inch cutting height, 20 minutes at 3-inch cutting height, and 20 minutes at 3/4- or 1-inch cutting height, to determine compliance with the applicable performance requirements of this specification.

4.4.2 Production unit tests. Each component part and each subassembly shall be assembly line tested as applicable and practicable to insure performance as required of the part, and of the completed mower. Not less than three cutting knives, selected at random from starting, midpoint and ending of each production run, shall be tested for hardness in accordance with ASTM E18. Failure of any knife to meet the specified hardness requirement shall be cause for testing every knife in production run. All defective knives shall be replaced with non defective knives. During assembly, the

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cutting height of each mower shall be adjusted in rotation at low, approximate midpoint and maximum cuting height, to insure presence and proper fit of all component parts. One wheel of each type I mower unit shall be lifted not less than 3 times without resultant change in adjustment. Not less than one out of each ten type III and type V mowers shall be tractor mounted; raised and lowered within the applicable range of positions; and pto driven for 1 minute at the constant rated speed of the pto drive, to determine dynamic balance, alinement of all working parts, and security of adjustment settings. Failure to pass any phase of this test shall be cause for rejection.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly and matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss, and to accomplish reduction in cube. Bolts, nuts, screws, pins and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking shall be printed on tags conforming to UU-T-81, type A. The tags and the printing thereon shall be waterproofed.

5.1.1.2 Preservatives. Preservatives shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unpainted surfaces. Unpainted exterior metal surfaces, including threaded surfaces and surfaces exposed by disassembly, shall be coated with type P-1 preservatives.

5.1.1.4 Engines. Engines, engine components and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type II, method I.

5.1.1.5 Dry type clutches. All surfaces of dry type clutch assemblies, except composition facings, shall be coated with a thin film of primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.6 Enclosed gears. Gear housings shall be filled to the operating level with lubricant conforming to the applicable lubrication order and actuated to insure coating of all interior parts and surfaces.

5.1.1.7 Drive chains. Exposed drive chains shall be coated with enough type P-21 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins and bushings. After the excess preservative has drained, the chains and sprockets shall be coated with type P-1 preservative.

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5.1.1.8 Drive belts and pulleys. Belts shall either be released from tension or shall be completely removed. Unpainted faces of pulley grooves shall be coated with primer conforming to TT-P-664 or MIL-P-46093. Belts removed shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.9 Power takeoff shaft and universal joints. The pto shaft shall be fully retracted. Unpainted, exposed machined surfaces of the pto shaft and universal joints shall be coated with type P-6 preservative. The universal joints and end connections shall be wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2. The wrap shall be secured in place with tape conforming to PPP-T-60, type IV.

5.1.1.10 Repair parts. Unless otherwise specified (see 6.2), repair parts shall be preserved and packaged in accordance with MIL-P-116. When specified (see 6.2), repair parts shall be preserved and packaged in accordance with level A of MIL-R-196, or where parts are not specifically covered in MIL-R-196, an applicable submethod of preservation in MIL-P-116 shall be used.

5.1.1.11 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40, level A.

5.1.1.12 Technical publications. Technical publications for each mower shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.13 Consolidated packaging.

5.1.1.13.1 Disassembled components, maintenance tools and technical publications. Small disassembled components, maintenance tools and technical publications shall be packed in close-fitting boxes conforming to PPP-B-601, domestic type. Large disassembled components, such as drawbars and connecting bars, shall be adaquately secured through bundling or other means to prevent loss. The contents of the boxes shall be cushioned, blocked and secured to prevent movement within the boxes. Strapping shall not be required.

5.1.1.13.2 Repair parts. The repair parts shall be packaged in a container as specified for disassembled components.

5.1.2 Level C. The complete mower shall be preserved and packaged in a manner that will insure adequate protection against deterioration and damage during shipment.

5.2 Packing. The packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The mowers shall be packed in open type crates conforming to MIL-C-52950, type V, or in boxes conforming to PPP-B-621, class 2. The contents shall be anchored, blocked and braced to prevent movement inside the containers or damaging of the contents.

5.2.2 Level B. The mowers shall be packed as specified for level A, except boxes shall conform to PPP-B-621, class 1.

5.2.3 Level C. The complete mowers shall be packed in a manner which will insure arrival at destination in satisifactory condition. Packing shall comply with applicable carrier rules and regulations.

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5.3 Marking. Interior packages and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Flail type mowers are intended for use in lieu of reel or rotary type mowers on uncleared, uncultivated, irregular land areas, fairways, parade gronds, terraces and lawns, where safety requires that trash, chip and clipping fly be minimized, and a large number of small replaceable cutter knives can insure continuous safe operation.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type, style and size required (see 1.2).
c. When first article is required for inspection and approval (see 3.2, 4.2.1 and 6.4).
d. Whether types III and type V mower shall be powered from a constant speed pto drive operating at 540 rpm or 1,000 rpm (see 3.9).
e. When heavy duty coarse cutting knives shall be provided (see 3.9.1).
f. Type of towing devices required (see 3.9.2).
g. When leaf mulching attachment is required (see 3.10).
h. Tractor on which type V mower shall be designed to operate (see 3.13).
i. When lubrication shall be other than manufacturer's standard practice (see 3.14).
j. When cleaning, treatment and painting shall conform to MIL-T-704 (see 3.15).
k. Color of finish coat required (see 3.15).
l. Level of preservation and level of packing required (see 5.1 and 5.2).
m. When repair parts are to be preserved and packaged in accordance with MIL-R-196 (see 5.1.1.10).

* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item consisting of one complete mower unit or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to

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evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - ME
Navy - YD
Air Force - 99

User Activity:

Army - CE

Preparing Activity:

Navy - YD

(Project 3750-0044)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-23421D	Mowers, Lawn; Flail Type, Towed and Tractor Mounting

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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